FARM MANAGEMENT SYSTEMS

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ABSTRACT

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THE FORM MANAGEMENT System is a web-based application developed to manage and organize agricultural information efficiently. The system helps store and maintain data related to farms, crops, farmers, and buyers in a structured and digital format. It reduces manual paperwork and provides an easy way to manage farm-related activities.

The system includes different modules such as Farms, Crops, Farmers, Buyers, Reports, and Dashboards, which help users easily access and update agricultural data. Through these modules, users can add new farm records, view crop information, maintain farmer details, and manage buyer transactions.The Farm Management System also provides features such as search options, data management, and report generation to analyze farm activities. These features help farm managers make better decisions regarding crop production and farm operations.

Overall, the system improves efficiency, accuracy, and data management in agricultural activities. It provides a simple and effective solution for monitoring and managing farm resources using modern technology.

INTRODUCTION

INTRODUTION

A Farm Management System is a digital platform used to manage and monitor different farming activities efficiently. The system helps farmers and agricultural organizations store, organize, and analyze important farm data such as farm details, crop information, farmers, buyers, and reports in one place.

The interface shown in the picture is developed using the cloud platform of Salesforce. It displays a dashboard where users can manage different modules such as Farms, Crops, Farmers, Buyers, Reports, and Dashboards. In the "Farms" section, the system shows a list of farms like Anandhan Farm and Yuvaraj Farm, which are recently viewed records.

This system improves farm productivity by allowing users to:

- *Maintain farm records digitally*
- *Track crop details and production*
- *Manage farmer and buyer information*
- *Generate reports for better decision making*
- *View data quickly through dashboards*

By using a Farm Management System, agricultural management becomes more organized, efficient, and data-driven. It reduces manual work and helps farmers and administrators make better decisions for improving farm performance

The Farm Management System is a digital platform used to manage agricultural data such as farm details, crops, farmers, and buyers. It helps store, organize, and analyze farm information efficiently

PROBLEM STATEMENT

PROBLEM STATEMENT

The agricultural sector still relies heavily on manual record keeping, which leads to inefficiency, data errors, and difficulty in managing farm-related information. Farmers and farm managers often maintain records of crops, farms, buyers, and production details in paper registers or scattered files. Because of this manual system, data is frequently incomplete, lost, or inaccurate

Farm managers face challenges in monitoring farm productivity and sales, as reports are often delayed and difficult to analyze. Additionally, buyers do not have access to updated crop availability and pricing information, which affects the transparency of agricultural transactions. System administrators also struggle to manage user access and maintain data security because access control is handled manually

Due to these issues, farmers and stakeholders experience frustration, inefficiency, and lack of proper decision-making tools. Therefore, there is a need for a centralized digital platform that can store, manage, and analyze farm data efficiently

The Farm Management System (FMS) aims to solve these problems by providing a Salesforce-based centralized system that allows farmers, managers, buyers, and administrators to record, monitor, and analyze agricultural data securely and efficiently

FEASIBILITY STUDY

FEASIBILITY STUDY

TECHNICAL FEASIBILITY:

The system is technically feasible because it is built using the Salesforce cloud platform, which provides secure data storage, scalability, and integrated tools for application development. The system uses Salesforce Lightning UI for the front-end and Salesforce cloud database for data storage, making it reliable and easy to maintain. Additionally, built-in features such as validation rules, reports, and dashboards support efficient farm data management

ECONOMIC FEASIBILITY:

The project is economically feasible because it reduces costs related to manual paperwork, data storage, and administrative efforts. Although Salesforce licensing may involve some cost, the benefits such as automation, improved productivity, and accurate data management outweigh the expenses. The system helps farmers and managers make better decisions, which can improve farm productivity and profitability

OPERATIONAL FEASIBILITY:

The Farm Management System is designed with a user-friendly interface, making it easy for farmers, farm managers, and administrators to operate. Users can easily enter farm data, manage crop records, and generate reports through a simple dashboard. The system also provides role-based access control, ensuring that users can only access relevant data. This improves operational efficiency and reduces errors

SCHEDULE FEASIBILITY:

The project can be completed within a planned timeframe using agile development methods with sprint schedules. Tasks such as system setup, data modeling, automation, security configuration, and reporting are divided into multiple sprints, ensuring that the system is developed and tested systematically

OBJECTIVE
OF THE SYSTEM

OBJECTIVE OF THE SYSTEM

The main objective of the Farm Management System is to help farmers and agricultural organizations manage farming activities in a simple, organized, and efficient way using a digital platform such as Salesforce

Objectives:

- To maintain farm records digitallyThe system stores information about farms, crops, farmers, and buyers in a centralized database.

- To improve farm management It helps manage daily farming activities such as crop monitoring, farm details, and production tracking.

- To manage farmer and buyer information It allows users to store and track details of farmers and buyers involved in agricultural activities.

- To generate reports and dashboards The system provides reports and visual dashboards to analyze farm performance and productivity.

- To improve decision making By analyzing stored data, farmers and administrators can make better decisions regarding crops, production, and sales.

- To increase efficiency and productivity The system saves time and helps manage agricultural resources more effectively

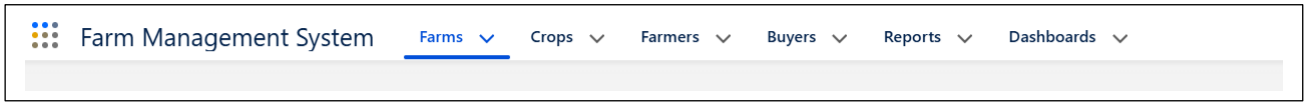
- These objectives help the Farm Management System improve agricultural operations, data organization, and overall farm productivity

SYSTEM OVERVIEW

SYSTEM OVERVIEW

The system is developed using Salesforce Lightning interface. It provides different modules such as:

- Farms
- Crops
- Farmers
- Buyers
- Reports
- Dashboards



MODULES
OF THE SYSTEM

MODULES OF THE SYSTEM

Modules of the System:

4.1 Farms Module

This module stores farm information such as:

- Farm Name
- Location
- Owner details
- Land size

Example records shown:

- Anandhan Farm
- Yuvaraj Farm

4.2 Crops Module

This module manages crop details such as:

- Crop name
- Crop type
- Season
- Quantity produced

4.3 Farmers Module

This module maintains farmer information including:

- Farmer name
- Contact details
- Farm ownership
- Farming activities

4.4 Buyers Module

This module manages buyers who purchase crops from farms.

Information stored:

- Buyer name
- Contact details
- Crop purchased
- Purchase quantity

4.5 Reports Module

Reports help generate summaries such as:

- Crop production reports
- Farm activity reports
- Buyer purchase reports

REPORTS

REPORTS

❖ FORMERS AND FORMS REPORT:

Farm Management System | Farms | Crops | Farmers | Buyers | Reports | Dashboards

Report: Farmers with Farms
Farmers with Farms Report

Enable Field Editing | Add Chart | Edit

Total Records
4

Farm Type	Farmer: Farmer Name	Farm: Farm Name	size	Soil Quality
Crop (3)	Yuvaraj	Yuvaraj Farm	15	Composting
	PACHAIYAPPAN RANGA	PACHAIYAPPAN FARM	10	Organic
	Arul moorthy	Arul Farm	15	Composting
Subtotal				
Mixed (1)	Anandhan	Anandhan Farm	10	Organic
Subtotal				
Total (4)				

❖ CROPS WITH FARM REPORTS:

Farm Management System | Farms | Crops | Farmers | Buyers | Reports | Dashboards

Report: Crops with Farm
Crops with Farm Report

Enable Field Editing | Add Chart | Edit

Total Records 5 | Total Actual Yield $1,100,000 | Total Total Acres Harvested 48 | Total Expected Yield per Acre $120,000.00

Farm: Farm Name	Crop: Crops Name	Actual Yield	Total Acres Harvested	Expected Yield per Acre	Farm: size	Farm: Farmer	Farm: Soil Quality
Anandhan Farm (2)	Cotton	$200,000	10	$20,000.00	10	Anandhan	Organic
	Rice	$300,000	15	$20,000.00	10	Anandhan	Organic
Subtotal		$500,000	25	$40,000.00			
Arul Farm (1)	Wheat	$400,000	8	$50,000.00	15	Arul moorthy	Composting
Subtotal		$400,000	8	$50,000.00			
PACHAIYAPPAN FARM (1)	Corn	$100,000	10	$10,000.00	10	PACHAIYAPPAN RANGA	Organic
Subtotal		$100,000	10	$10,000.00			
Yuvaraj Farm (1)	Coconet	$100,000	5	$20,000.00	15	Yuvaraj	Composting
Subtotal		$100,000	5	$20,000.00			
Total (5)		$1,100,000	48	$120,000.00			

❖ Farm Details and Acres:

Previewing a limited number of records. Run the report to see everything.

	Crop: Crops Name	Farm: Farm Name	Total Acres Harvested
1	Rice	Anandhan Farm	15
2	Cotton	Anandhan Farm	10
3	Coconet	Yuvaraj Farm	5
4	Corn	PACHAIYAPPAN FARM	10
5	Wheat	Arul Farm	8
6			48

❖ Farmers Size and Crop Name:

Previewing a limited number of records. Run the report to see everything.

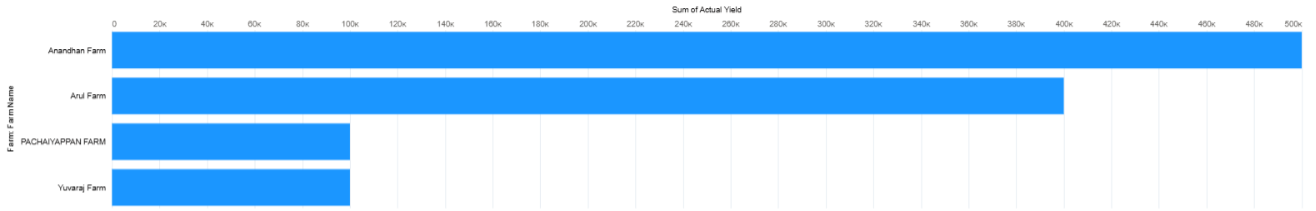
	Crop: Crops Name	Farm: Farm Name	Farm: size	Actual Yield
1	Rice	Anandhan Farm	10	$300,000
2	Cotton	Anandhan Farm	10	$200,000
3	Coconet	Yuvaraj Farm	15	$100,000
4	Corn	PACHAIYAPPAN FARM	10	$100,000
5	Wheat	Arul Farm	15	$400,000
6				$1,100,000

❖ Farmers Details:

Previewing a limited number of records. Run the report to see everything.

	Farmer: Farmer Name	size	Soil Quality	State	District	Village	Farm Type	Irrigation type	Contact Number	Farmer: ID
1	Anandhan	10	Organic	Tamil nadu	Chennai	Kolathur	Mixed	Drip Irrigation	9360438824	a02fj00000Rgbx6
2	Yuvaraj	15	Composting	Tamil nadu	Coimbatore	Bellapalayam	Crop	Local irrigation	9876789765	a02fj00000RgZIG
3	PACHAIYAPPAN RANGA	10	Organic	Telangana	Bhongir	Anakapelle	Crop	Drip Irrigation	8978907895	a02fj00000RpFEP
4	Arul moorthy	15	Composting	Tamil nadu	Coimbatore	Ezhichur	Crop	Drip Irrigation	9879657890	a02fj00000Roocc
5	Dharma	10	Composting	Tamil nadu	Chennai	Cheemalapadu	Crop	Sprinkler irrigation	9879756789	a02fj00000RpyEZ

❖ Farmers Details With Chart Table:



Farm: Farm Name	Crop: Crops Name	Actual Yield	Total Acres Harvested	Expected Yield per Acre	Farm: size	Farm: Farmer	Farm: Soil Quality
Anandhan Farm (2)	Cotton	$200,000	10	$20,000.00	10	Anandhan	Organic
	Rice	$300,000	15	$20,000.00	10	Anandhan	Organic
Subtotal		$500,000	25	$40,000.00			
Arul Farm (1)	Wheat	$400,000	8	$50,000.00	15	Arul moorthy	Composting
Subtotal		$400,000	8	$50,000.00			
PACHAIYAPPAN FARM (1)	Corn	$100,000	10	$10,000.00	10	PACHAIYAPPAN RANGA	Organic
Subtotal		$100,000	10	$10,000.00			
Yuvaraj Farm (1)	Coconet	$100,000	5	$20,000.00	15	Yuvaraj	Composting
Subtotal		$100,000	5	$20,000.00			
Total (5)		$1,100,000	48	$120,000.00			

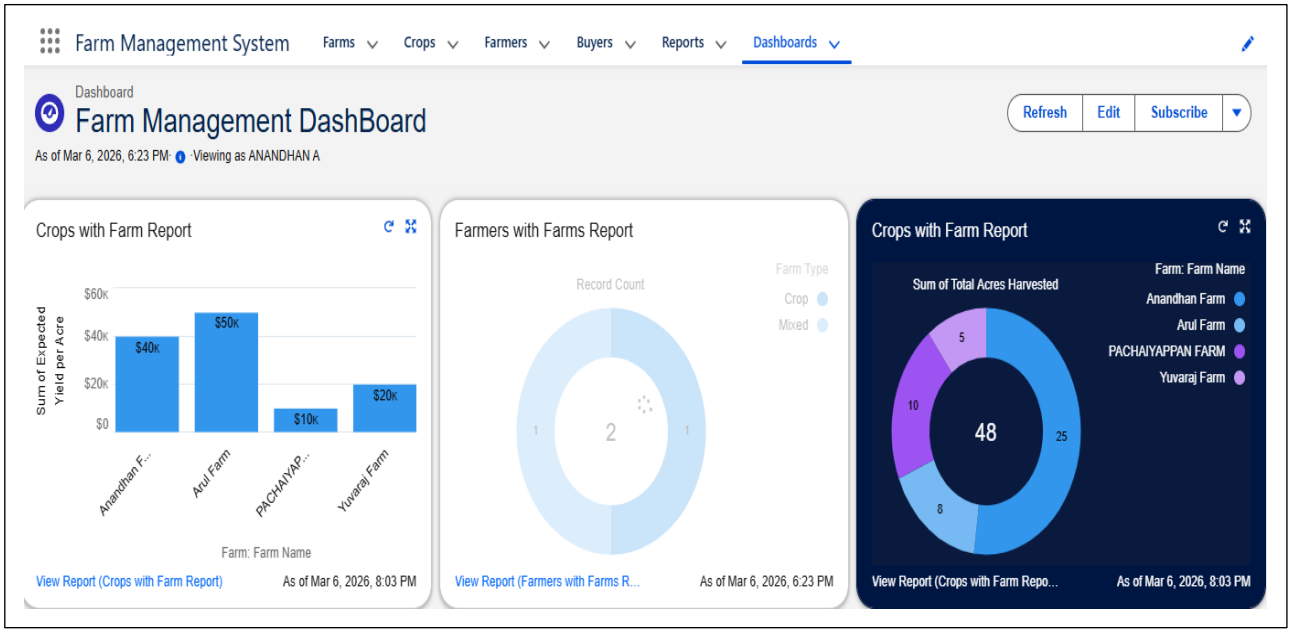
DASHBOARD

DASHBOARD

COMPONENETS:

- Bar Chart- Sum Of Expected Yield per Acre
- Pie chart- Record Type in Crop And Mixed Farm Type
- Pie Chart- Sum Of Total Acres Hevested

Dash Board explained a Overall Farm Management Analytics:



SYSTEM ARCHITECTURE

SYSTEM ARCHITECTURE

The FARM MANAGEMENT System Have Four Main Custom Object:

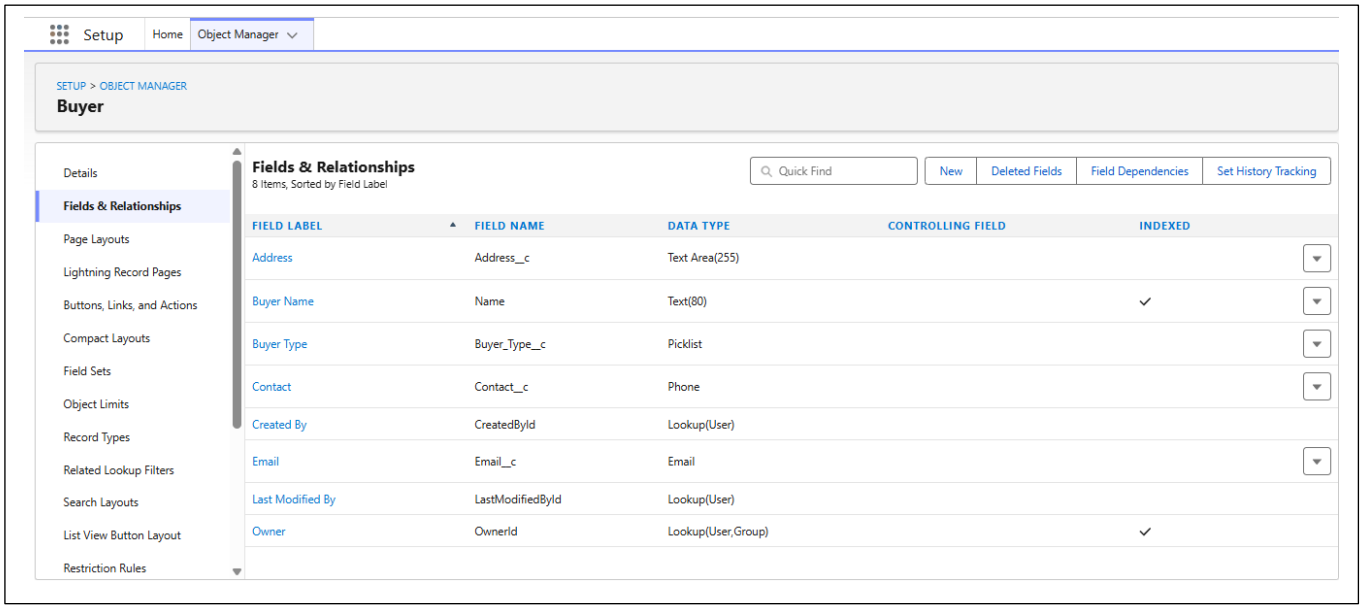
1.Buyer

2.Crop

3.Farm

4.Farmer

❖ Buyer:



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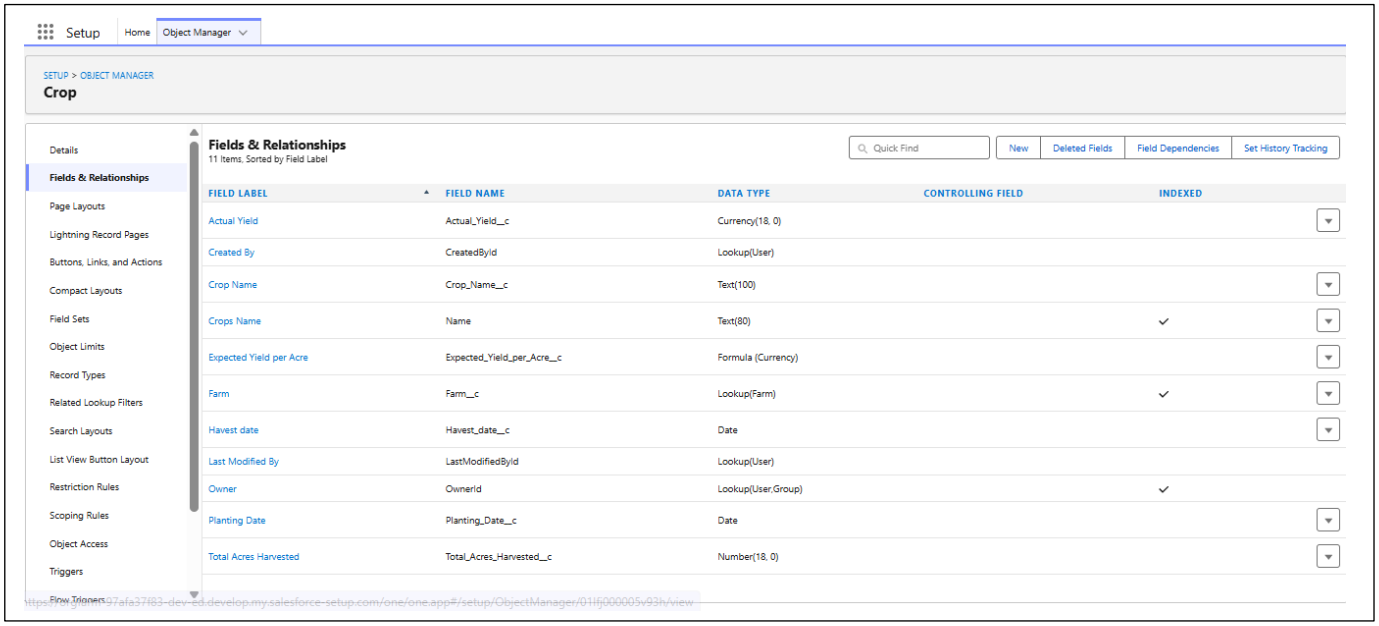
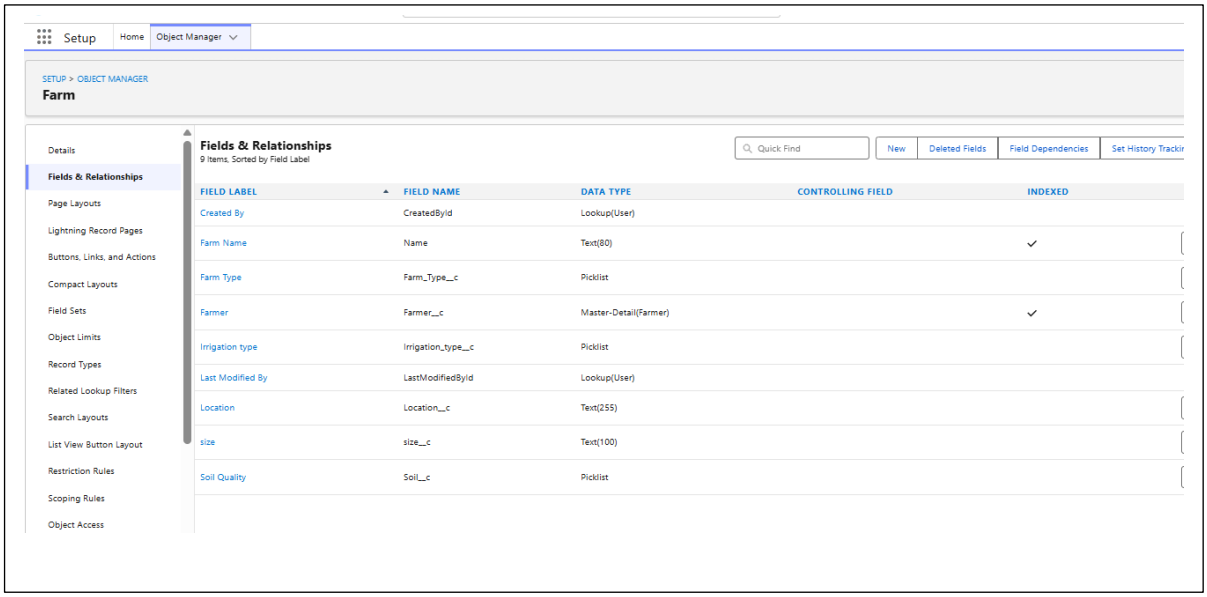

Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Crop
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Scoping Rules
Object Access
Triggers
Flow Triggers
Fields & Relationships
11 Items, Sorted by Field Label
Quick Find
New
Deleted Fields
Field Dependencies
Set History Tracking
FIELD LABEL
FIELD NAME
DATA TYPE
CONTROLLING FIELD
INDEXED
Actual Yield
Actual_Yield__c
Currency(18, 0)
Created By
CreatedById
Lookup(User)
Crop Name
Crop_Name__c
Text(100)
Crops Name
Name
Text(80)
Expected Yield per Acre
Expected_Yield_per_Acre__c
Formula (Currency)
Farm
Farm__c
Lookup(Farm)
Havest date
Havest_date__c
Date
Last Modified By
LastModifiedById
Lookup(User)
Owner
OwnerId
Lookup(User,Group)
Planting Date
Planting_Date__c
Date
Total Acres Harvested
Total_Acres_Harvested__c
Number(18, 0)

❖ Farm:


Setup
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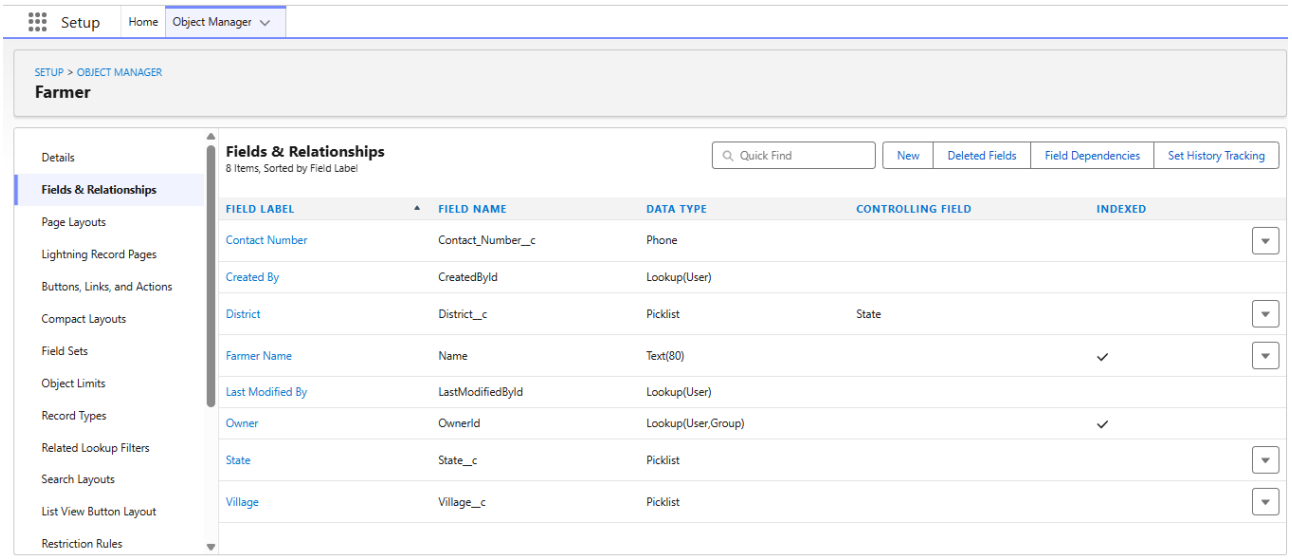
SETUP > OBJECT MANAGER
Farm
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Scoping Rules
Object Access
Fields & Relationships
9 Items, Sorted by Field Label
Quick Find
New
Deleted Fields
Field Dependencies
FIELD LABEL
FIELD NAME
DATA TYPE
CONTROLLING FIELD
INDEXED
Created By
CreatedById
Lookup(User)
Farm Name
Name
Text(80)
Farm Type
Farm_Type__c
Picklist
Farmer
Farmer__c
Master-Detail(Farmer)
Irrigation type
Irrigation_type__c
Picklist
Last Modified By
LastModifiedById
Lookup(User)
Location
Location__c
Text(255)
size
size__c
Text(100)
Soil Quality
Soil__c
Picklist

❖ Farmers:



FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Contact Number	Contact_Number__c	Phone		
Created By	CreatedById	Lookup(User)		
District	District__c	Picklist	State	
Farmer Name	Name	Text(80)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
State	State__c	Picklist		
Village	Village__c	Picklist		

DATA MODEL DESIGN

DATA MODEL DESIGN

The Farms object Stores personal Information:

- Farm Name
- Farm Type
- Soil Quality
- Irrigation Type
- Location
- Size
- Former

The Crops Object Stores Information:

- Crops name
- Planting Date
- Havest Date
- Farm
- Actual Yield
- Total Acres Havested

The Former Object Stores Information:

- Farmer Name
- Contact Name
- State
- District
- Village

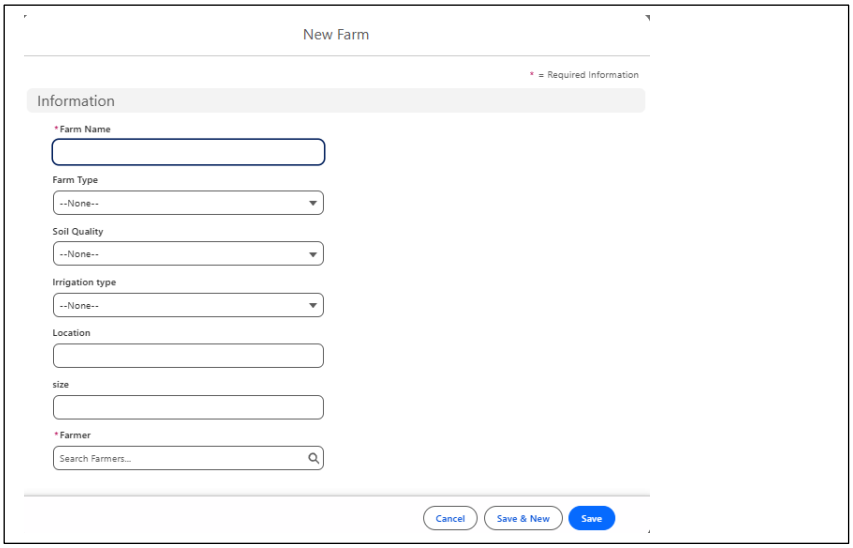
The Buyer Object Stores Information:

- Buyer Name
- Cantact
- Email
- Address
- Buyer Type

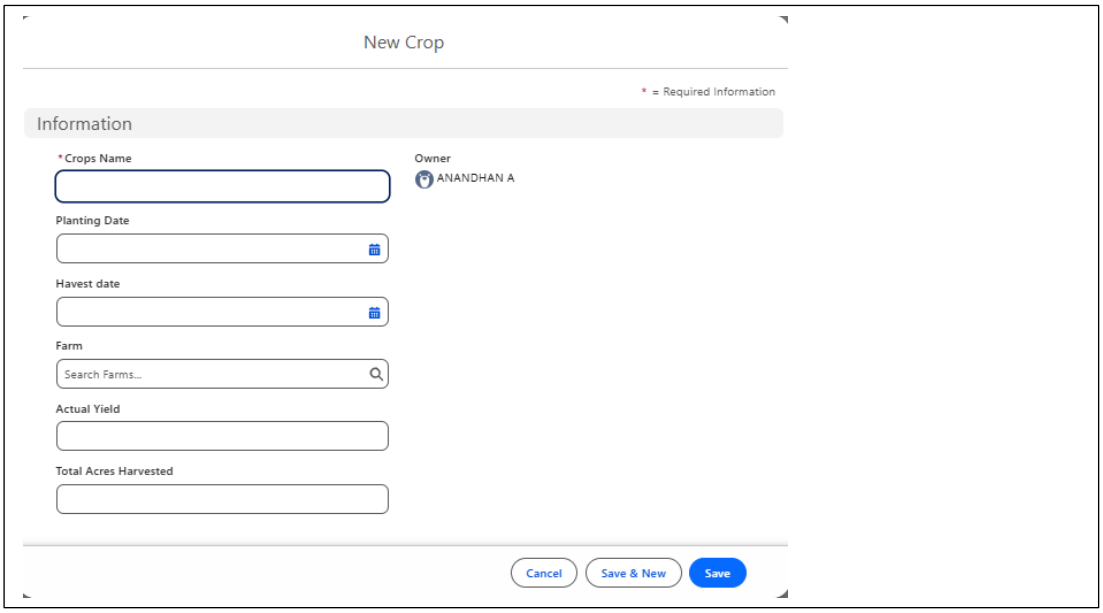
MODULE INPUT,OUTPUT SCREENS

MODULE INPUT SCREENS

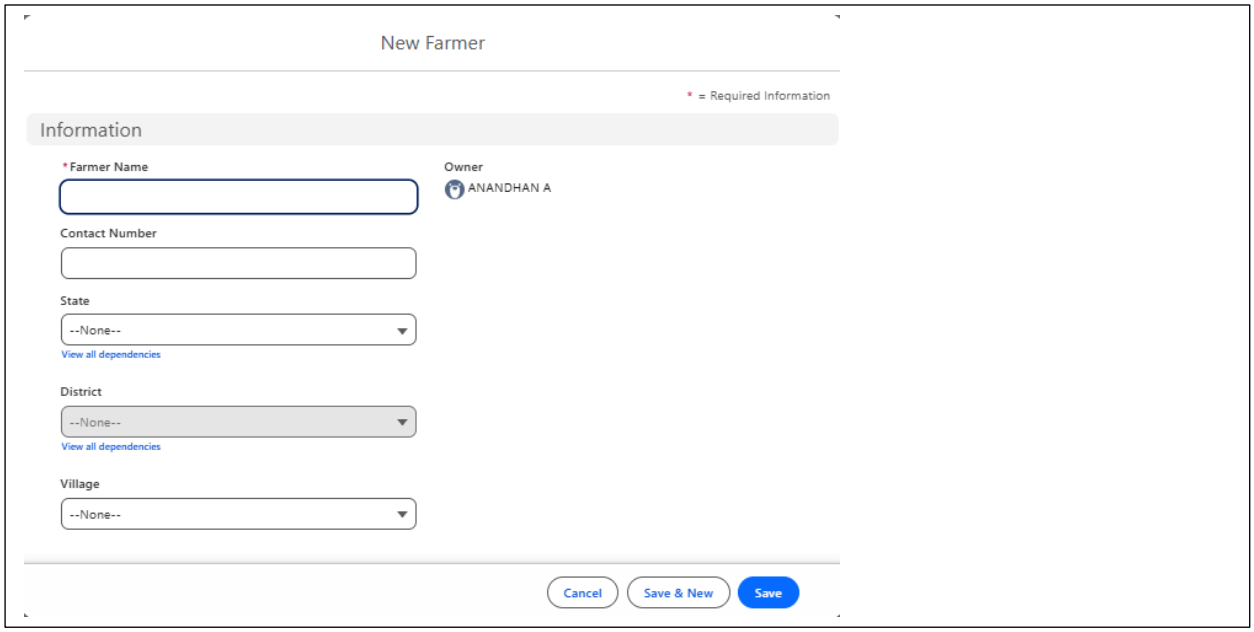
New Farm Creation:

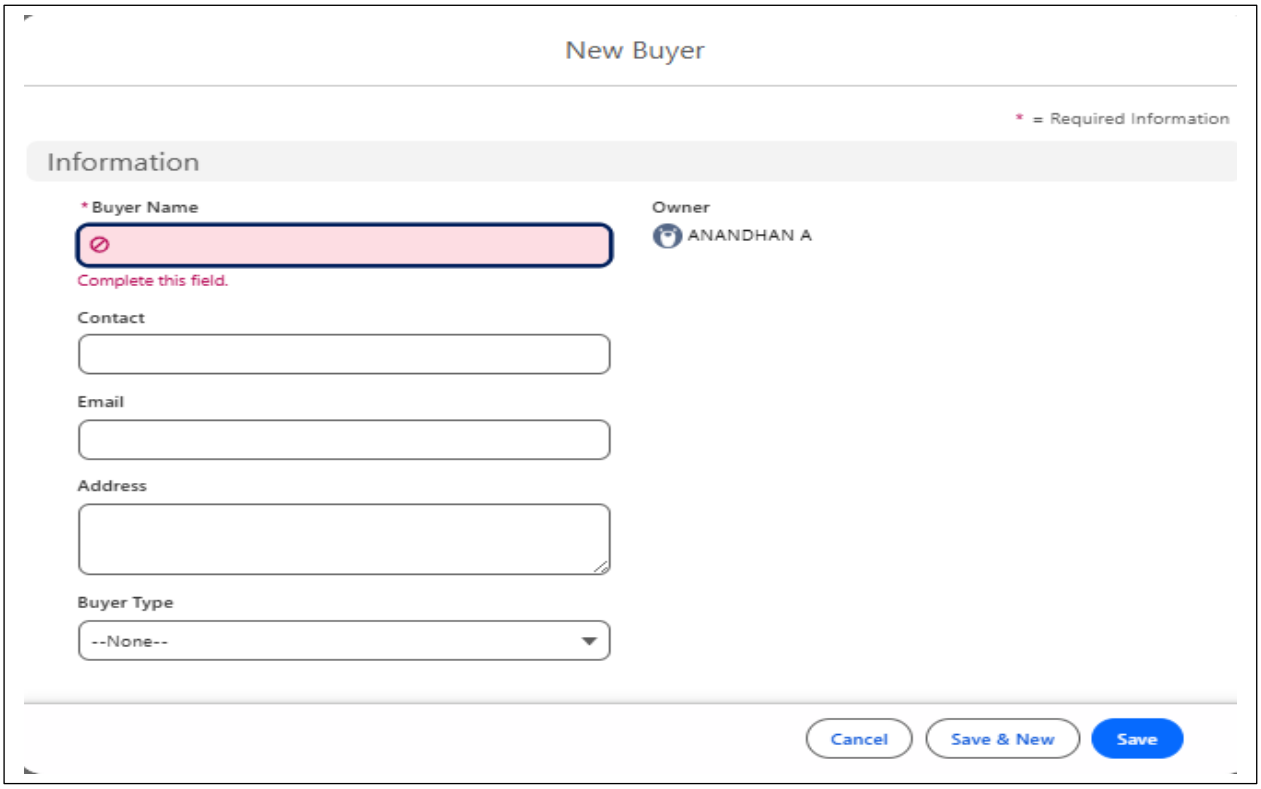


New Crop Creation:



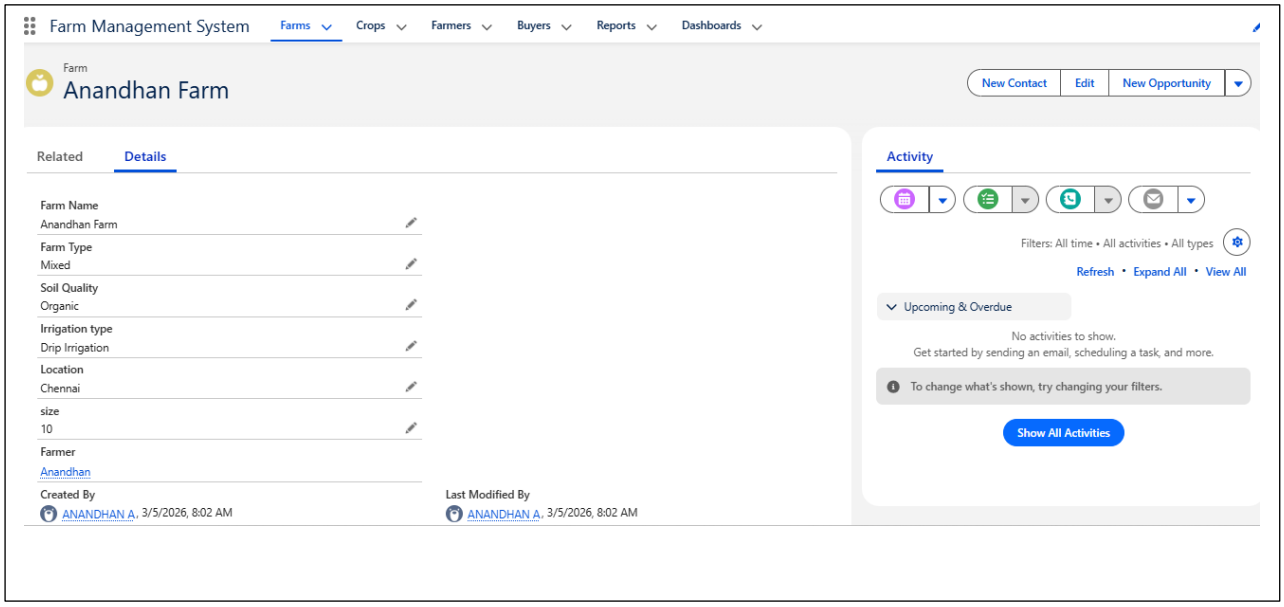
New Farmer Creation:



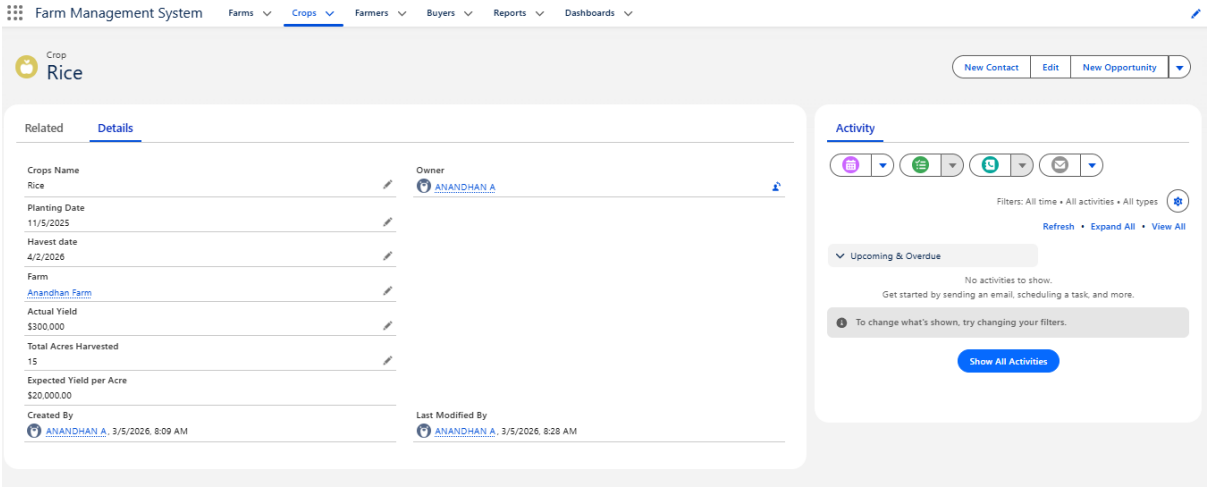


Output Screen:

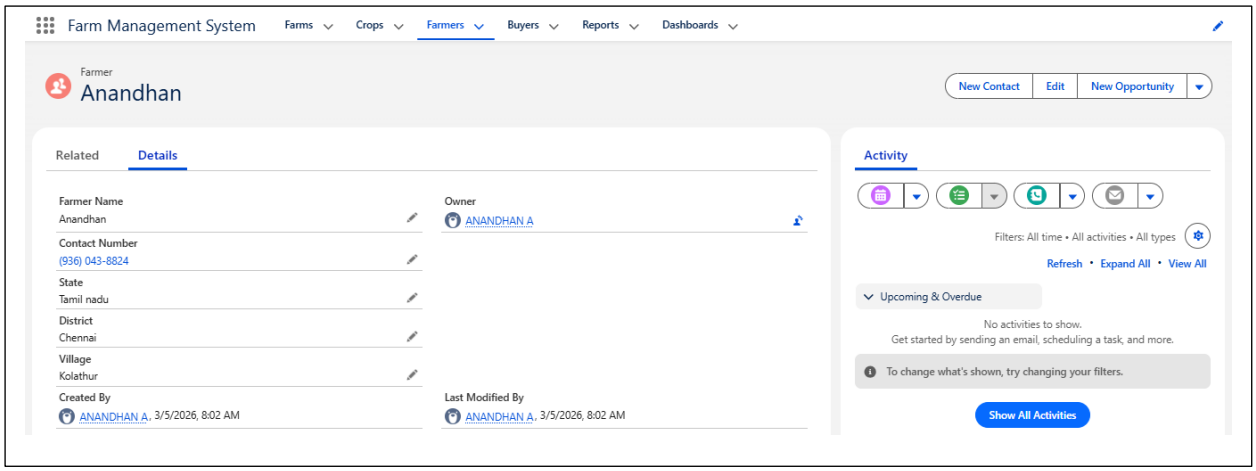
Farm Output:



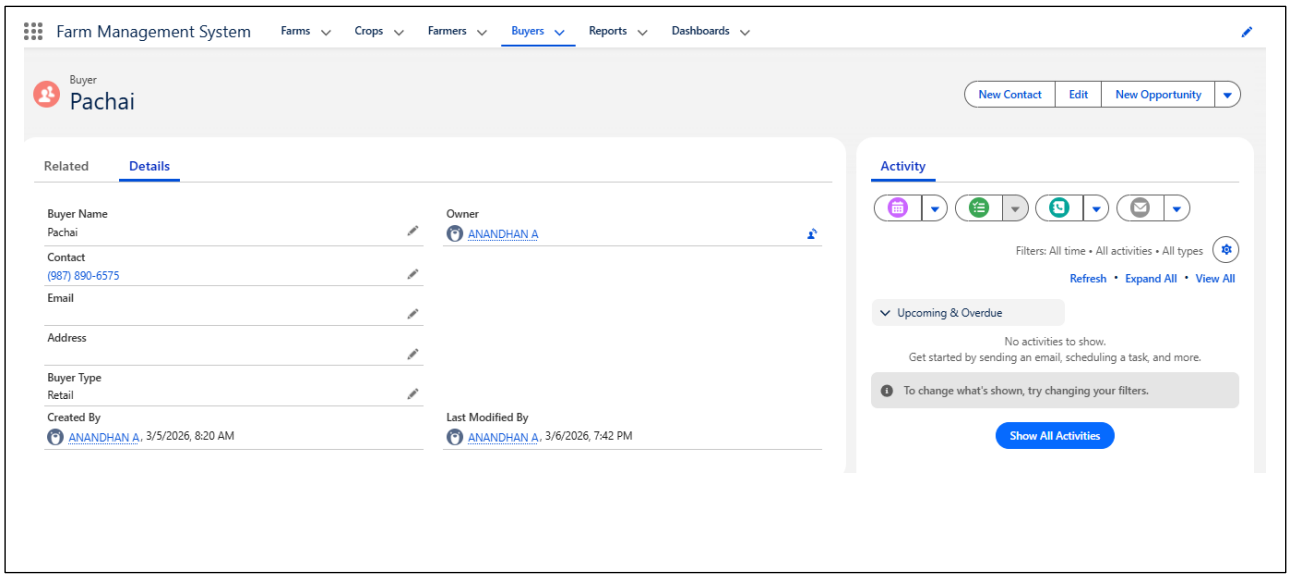
Crop Output:



Farmers Output:



Buyers Output:



ADVANTAGES OF FARM MANAGEMENT SYSTEM

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Centralized Data Management:

All farm-related information such as farm details, crop records, farmer data, and buyer information are stored in a single system, making it easy to manage and access.

Reduced Paperwork:

The system replaces manual registers and paper records with digital data storage, reducing paperwork and saving time.

Improved Data Accuracy:

Validation rules and automated data entry help prevent incorrect or incomplete information, improving the accuracy of records.

Real-Time Monitoring:

Farmers and farm managers can monitor crop production, farm activities, and sales in real time using reports and dashboards.

Better Decision Making:

The system provides analytical reports and dashboards that help users analyze farm productivity and make informed decisions.

Secure Data Storage:

Role-based access control ensures that only authorized users can access sensitive farm data, improving security.

Easy Data Access:

Since the system is cloud-based, users can access farm data anytime and from anywhere with an internet connection.

Improved Farm Productivity:

By organizing farm operations efficiently, the system helps farmers increase productivity and manage resources better.

Faster Report Generation:

Reports and dashboards are generated automatically, saving time compared to manual calculations.

Better Collaboration:

Farmers, managers, buyers, and administrators can work together using the same platform, improving communication and coordination

CONCLUSION

CONCLUSION

The Farm Management System (FMS) successfully provides a digital solution for managing agricultural data and farm operations. By replacing manual record-keeping with an automated system, the project improves efficiency, reduces errors, and ensures better organization of farm-related information.

The use of the Salesforce platform allows the system to store data securely in the cloud while providing real-time access to farmers, farm managers, and administrators. Features such as user authentication, role-based access control, validation rules, reports, and dashboards enhance the overall functionality of the system.

Through this project, farmers and stakeholders can easily manage farms, crops, farmers, and buyer information in a centralized system. The system also supports better decision-making by providing accurate data and visual insights into farm productivity and sales performance.

In conclusion, the Farm Management System contributes to the modernization of agricultural management by integrating technology into traditional farming practices. With further enhancements such as mobile applications, AI-based crop prediction, and integration with weather and IoT systems, the platform has the potential to become an advanced digital solution for smart farming in the future.